

Leo Wang

Email: leokingw@andrew.cmu.edu | Cell: (860) 960-9539 | Portfolio: leo01110111.github.io/projects

EDUCATION

Carnegie Mellon University | Pittsburgh, PA

B.S. Electrical and Computer Engineering | May 2028

GPA: 4.0/4.0

The Hotchkiss School | Lakeville, CT

High School Diploma | May 2024 | GPA 3.9/4.0

PROJECT EXPERIENCE

Quadrupedal Climbing via Reinforcement Learning | May 2024 - Present

- Developed a reinforcement learning framework in NVIDIA Isaac Lab to train a quadrupedal robot to climb hilly terrain.

Robotic Arm | Personal Project | June 2024 - Present

- Made a 6-degree-of-freedom robotic arm from the ground up using 3D printed parts
- Wrote Arduino software to control the arm with inverse kinematics

Drawing Robot | Personal Project | Sep 2024 - Present

- Constructed a robot that draws on walls using 3D printed parts, stepper motors, and a servo motor
- Coded an program that enables the robot to reproduce digital images on walls

Robot Dog | Group Project | Oct 2024 - Present

- Designed carbon fiber body using Solidworks
- Familiarized with common components in robotic projects: Orin Nano, cycloidal gearboxes, BLDC motors

WORK EXPERIENCE

Robomechanics Lab | Researcher | Carnegie Mellon University | May 2025 - Present

- Working on a reinforcement learning controller for a robotic quadruped to climb steep cliffs

Bio-Robotics Lab | Intern | Chinese University of Hong Kong | June - September 2023

- Manufactured a prosthetic for an arm amputee, allowing her to play the violin
- Built 6 iterations of the prosthetic until it was lighter than 100g and comfortable to wear

LEADERSHIP EXPERIENCE

Hotchkiss Robotics Team (FIRST) | Captain, Mechanical Designer | 2021 - 2024

- Oversaw hardware, software, outreach, and logistical projects
- Created a mechanical subsystem for the robot using CAD, 3D printing, and laser cutting
- Competed in the world championship

Engineering Teacher | 2022-2024

- Created a robotics class taught to 100 kids at 4 different elementary schools
- Taught a 6 week course on building a model city. 25 kids enrolled

Maker's Club | Founder and Head | 2022-2024

- Provided mentorship in the makerspace and organized engineering competitions

Computer Science Club | Head | 2022-2024

- Held coding workshops, discussions on CS related topics, and coding competitions

The St. Baldrick's Foundation | Event Organizer | 2018 - 2024

- Organized yearly fundraiser at school for childhood cancer research. Raised \$30,000

SKILLS

Hardware: CAD in Solidworks and Onshape, GD&T, FEA in Ansys Discovery, Arduino Prototyping

Machines: 3D Printer, Laser Cutter, Drill, Jigsaw, Bandsaw.

Software: Isaac Lab, MATLAB, Python, C, Java

ACTIVITIES AND HONORS

- CMU Robotics Club | September 2024 - Present
- Hotchkiss Robotics Team | 2021-2024
- First Honor Roll, 7 semesters | 2020-2024